

TAEJUN HA

tjha@yonsei.ac.kr

 Google Scholar |  GitHub |  ORCID

RESEARCH INTERESTS

Machine learning for physical-layer wireless communications: model-based deep learning for channel estimation, attention transformers and generative models (e.g. diffusion) as learned signal priors, joint channel estimation and decoding, and low-complexity receiver design under limited pilots and imperfect training labels. Broader interests include MIMO transceiver optimization and physical-layer design for AI-native 6G and virtualized RAN.

EDUCATION

- **Yonsei University** Mar 2025 – Feb 2027
M.S. in Electrical and Electronic Engineering (expected) Seoul, South Korea
 - Advisor: Prof. Jeonghun Park
 - Thesis: Learning Linear Filters for OFDM Channel Estimation Using an Attention Transformer
- **Yonsei University** Mar 2019 – Feb 2025
B.S. in Electrical and Electronic Engineering Seoul, South Korea

PUBLICATIONS AND PATENTS

J=JOURNAL, C=CONFERENCE, P=PATENT

- [J.1] T. Ha, H. Kim, and J. Park, "Attention-Aided MMSE with Ridge Denoising: How to Train under Noisy Channel Samples," *IEEE Transactions on Vehicular Technology*, accepted for publication, 2026. [[Arxiv](#)]
- [J.2] T. Ha, C. Jung, H. Kim, J. Park, and J. Park, "Learning MMSE Filters for OFDM Channel Estimation: Attention Transformer Gains at Linear Inference," submitted to *IEEE Transactions on Wireless Communications*. [[Arxiv](#)] (Under Review, Minor Revision)
- [J.3] T. Ha and J. Park, "Learning Linear Filters for Underwater OFDM Channel Estimation with Attention-Aided MMSE," *Journal of the Korean Institute of Communications and Information Sciences (Letters)*, 2026. (Published)
- [C.1] T. Ha and J. Park, "Attention-Aided MMSE for OFDM Channel Estimation," in *Proc. 59th Asilomar Conf. Signals, Systems, and Computers*, Pacific Grove, CA, USA, 2025.
- [P.1] J. Park and T. Ha, "Channel Estimation System and Method Using A-MMSE in OFDM Systems," Korean Patent Application No. 10-2025-0083228, filed Jun. 24, 2025. Applicant: Defense Acquisition Program Administration, Republic of Korea. (pending)
- [P.2] J. Park and T. Ha, "Method and Apparatus for Channel Estimation Using an Adaptive Channel-Estimation Filter," Korean Patent Application No. 10-2026-0127455, filed Jul. 10, 2026. Applicant: Yonsei University Industry-Academic Cooperation Foundation. (pending)

RESEARCH EXPERIENCE

- **The University of Texas at Austin** Jul 2026 – Dec 2026
Visiting Scholar, Dept. of Electrical and Computer Engineering Austin, TX, USA
 - Host: Prof. Hyeji Kim
 - Research topic: Diffusion-based joint channel estimation and decoding for pilot-limited MIMO.
 - Method: Learned diffusion prior for the channel, BCJR decoder as an exact symbol denoiser, coupled by extrinsic message passing.
 - Developed a ridge-regularized training objective and covariance-shrinkage surrogate targets that allow attention-based channel estimators to be trained without clean CSI labels, closing most of the gap to clean-label training at high SNR [J.1].

• **Yonsei University**

Mar 2025 – Present

Seoul, South Korea

Graduate Research Assistant, Advisor: Prof. Jeonghun Park

- **6GARROW: 6G AI-Native Integrated RAN-Core Networks** (EU Horizon Europe SNS JU, Grant No. 101192194; with IITP, Korea) — Designed an attention-transformer channel estimator for 6G OFDM receivers that attains near-MMSE accuracy at substantially lower computational cost, without requiring full second-order channel statistics.
- **Development of Intelligent Secure Underwater Communication Technology** (funded by KRIT, Korea) — Extended the attention-transformer estimator to underwater acoustic OFDM channels, improving estimation accuracy and link reliability under long delay spread and severe Doppler, supporting secure and dependable underwater links.
- Formulated the attention-aided MMSE (A-MMSE) framework, which learns linear channel-estimation filters via an Attention Transformer and performs inference through a single linear operation, removing the need for exact channel covariance; developed a two-stage Attention encoder and a rank-adaptive extension for a flexible performance-complexity trade-off. Presented at **Asilomar 2025** [C.1] and extended into a journal version **under minor revision at IEEE Trans. Wireless Communications** [J.2].

GRADUATE COURSEWORK

Advanced Multi-Antenna System Design and Analysis, Numerical Optimization, Information Theory, Broadband Wireless Communication Systems, Random Processes, Computer Vision, Network System Analysis, Advanced Programming.

TECHNICAL SKILLS

- **Programming:** Python, MATLAB
- **Frameworks & Tools:** PyTorch, NumPy/SciPy, Git, $\LaTeX$
- **Domain-Specific:** OFDM/MIMO link-level simulation; standardized channel models (COST 2100, 3GPP TDL/CDL); LMMSE and model-based channel estimation; attention transformers and diffusion models for physical-layer design; joint detection and decoding (BCJR, message passing)

HONORS AND AWARDS

- **BK21 FOUR Graduate Fellowship**, Ministry of Education / NRF, Korea 2025
- **DB Scholarship for Outstanding Incoming Students**, Yonsei University 2025
- **Jinri Academic Excellence Scholarship**, Yonsei University (3 semesters) 2022 – 2024